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System and method for testing durability of gas diffusion layer (GDL)

Abstract

A system for testing durability of a gas diffusion layer (GDL) is provided, including a clamping component, a liquid storage tank and a peristaltic pump. The clamping component is configured to fix the GDL. The liquid storage tank is configured to store a corrosive liquid. The peristaltic pump is arranged between the clamping component and the liquid storage tank. The peristaltic pump is connected to the clamping component via a first pipeline, and is connected to the liquid storage tank via a second pipeline. A GDL durability testing method is also provided. During the testing, the corrosive liquid is pumped into the clamping component to perform erosion-corrosion on the GDL.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority from Chinese Patent Application No. 202410643655.4, filed on May 23, 2024. The content of the aforementioned application, including any intervening amendments made thereto, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) This application relates to fuel cells, and more particularly to a system and method for testing durability of a gas diffusion layer (GDL).

BACKGROUND

(3) The gas diffusion layer (GDL) is an important component of the proton exchange membrane

fuel cell (PEMFC), which is located between the flow field and the catalyst layer. It plays an important role in supporting the catalyst layer and collecting the current, and simultaneously provides multiple channels for gas, protons, electrons, and water to participate in the electrode reactions. Generally, the GDL is fabricated by coating a microporous layer (MPL) on a gas diffusion backing (GDB) composed of a porous carbon substrate, such as carbon cloth, carbon nonwoven fabric and carbon paper.

(4) Under normal working conditions of the PEMFC, due to the simultaneous occurrence of chemical and physical changes and material transfer, the membrane electrode is required to be simultaneously responsible for the heat transfer, mass transfer, electrical conduction and structural support. Therefore, in addition to the evaluation of the maximum performance, the evaluation of durability is also crucial in the process of promoting the commercialization of PEMFC.

(5) Traditional PEMFC durability test methods mostly focus on the proton conductivity of the proton exchange membrane (PEM) and catalyst attenuation, and less attention has been paid to changes in the GDL. Although the GDL attenuation will not directly cause the PEMFC failure, a highly-durable GDL is indispensable for the long-term stable operation of the PEMFC.

(6) For the online testing of the PEMFC durability after the stack assembly, the testing results are often greatly affected by the PEM and catalyst. In this case, it is not easy to extract the GDL attenuation results from the online test results, making it difficult to accurately evaluate the GDL.

SUMMARY

(7) An object of the disclosure is to provide a system and method for testing durability of a gas diffusion layer (GDL).

(8) In order to achieve the above object, the following technical solutions are adopted.

(9) In a first aspect, this application provides a system for testing durability of a GDL, comprising: a clamping component; a liquid storage tank; and a peristaltic pump; wherein the clamping component is configured to fix the GDL, a middle of the clamping component is provided with a flow channel, and both ends of the clamping component are channel-free; the liquid storage tank is configured to store a corrosive liquid; the peristaltic pump is arranged between the clamping component and the liquid storage tank; and the peristaltic pump is connected to the clamping component via a first pipeline, and is connected to the liquid storage tank via a second pipeline; and the peristaltic pump is configured to be turned on to pump the corrosive liquid from the liquid storage tank into the clamping component during testing, so as to perform erosion-corrosion on the GDL.

(10) In some embodiments, the clamping component comprises a plurality of clamping plates arranged in parallel; and each of the plurality of clamping plates is provided with a water inlet and a water outlet; the water inlet is connected to the first pipeline, and the water outlet is connected to a third pipeline connected to the liquid storage tank; and the water inlet and the water outlet are configured such that the corrosive liquid enters the flow channel through the water inlet to perform the erosion-corrosion on the GDL, and flows back to the liquid storage tank through the water outlet.

(11) In some embodiments, a front end of the water inlet is connected to a float-type flowmeter and a pulse valve in sequence, and the pulse valve is connected to the peristaltic pump.

(12) In some embodiments, the plurality of clamping plates have the same size, and are each provided with a positioning hole at the same position; and the plurality of clamping plates are configured to be fixed via a screw passing through the positioning hole.

(13) In some embodiments, the plurality of clamping plates includes a combination of at least two of a blind plate without flow channels on both sides, an end plate with one flow-channel side surface, and a middle plate with two flow-channel side surfaces.

(14) In some embodiments, the GDL is located at a middle of the two adjacent clamping plates, and a gasket is provided around the GDL for sealing.

(15) In some embodiments, a heating component is provided on an outer side of the clamping

component.

(16) In some embodiments, the corrosive liquid is selected from the group consisting of water, Fenton's reagent, ethanol, acetone, hydrogen peroxide, sulfuric acid, hydrochloric acid, nitric acid, sodium hydroxide and a combination thereof.

(17) In some embodiments, the liquid storage tank comprises a blue-capped mobile phase bottle with a height of 100-300 mm.

(18) In a second aspect, this application provides a method for testing GDL durability using the above system, comprising: (1) providing a plurality of GDLs, wherein the plurality of GDLs are identical to each other; (2) respectively clamping the plurality of GDLs between the plurality of clamping plates; (3) turning on the peristaltic pump to perform erosion-corrosion on the plurality of GDLs, and taking out N GDLs among the plurality of GDLs every M days; and (4) determining the physical and chemical parameters of the plurality of GDLs and an uncorroded GDL by using an analytical instrument.

(19) Compared to the prior art, the present disclosure has the following beneficial effects.

(20) The present disclosure can perform the offline GDL durability test under the corrosion of the corrosive liquid.

(21) Compared with the conventional immersion, the system provided herein can simulate the influence of water and gas flushing in the flow channel of the cell stack and the corrosive liquid on a microporous layer (MPL) and a gas diffusion backing (GDB) layer in the GDL.

(22) For the system of the present disclosure, the corrosive liquid (such as oxidizing solution, acidic solution, etc.) can be readily replaced. An appropriate combination of end plates, middle plates and blind plates can be adopted to clamp different numbers of samples and flush different surfaces of the samples. The gasket thickness is adjustable, such that the compression ratio of the GDL can be changed by changing the gasket thickness, thereby simulating the corrosion resistance of the GDL under different assembly pressures.

(23) A pulse valve is provided at the inlet of the corrosive liquid, so as to simulate a non-steady flow under an operating condition of a fuel cell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a structural diagram of a system for testing durability of a gas diffusion layer (GDL) in accordance with an embodiment of the present disclosure;

(2) FIG. 2 is a structural diagram of a clamping component in accordance with an embodiment of the present disclosure;

(3) FIG. 3 is a structural diagram of a clamping component in accordance with Embodiment 1 of the present disclosure;

(4) FIG. 4 shows weight loss rate of the GDL in accordance with an embodiment of the present disclosure;

(5) FIG. 5 shows surface resistivity of the GDL in accordance with an embodiment of the present disclosure; and

(6) FIG. 6 shows machine-direction (MD) stiffness of the GDL in accordance with an embodiment of the present disclosure.

(7) In the drawings, **100.** clamping component; **110.** first water inlet; **120.** first water outlet; **130.** clamping plate; **141.** front end plate; **142.** first middle plate; **143.** second middle plate; **144.** third middle plate; **145.** fourth middle plate; **146.** fifth middle plate; **147.** sixth middle plate; **148.** seventh middle plate; **149.** eighth middle plate; **1410.** ninth middle plate; **1411.** rear end plate; **140.** positioning hole; **150.** screw; **160.** thermocouple; **170.** gasket; **200.** liquid storage tank; **210.** second water inlet; **220.** second water outlet; **230.** water bath; **300.** peristaltic pump; **400.** float-type

flowmeter; **500**. pulse valve; **600**. heating component; **700**. filtering tank; and **800**. GDL.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) In order to better illustrate the objects, technical solutions and advantages of the present disclosure, the present disclosure will be further described below in conjunction with the embodiments.

(9) In the present disclosure, unless otherwise specified, the test methods are all conventional methods. Materials and reagents used herein are all commercially available unless otherwise specified.

(10) As shown in FIG. 1, a system for testing durability of a gas diffusion layer (GDL) is provided, which includes a clamping component **100**, a liquid storage tank **200** and a peristaltic pump **300**. The clamping component **100** is configured to fix the GDL **800**. The liquid storage tank **200** is configured to store a corrosive liquid. The peristaltic pump **300** is arranged between the clamping component **100** and the liquid storage tank **200**. The peristaltic pump **300** is connected to the clamping component **100** via a first pipeline, and is connected to the liquid storage tank **200** via a second pipeline.

(11) The peristaltic pump **300** is configured to be turned on to pump the corrosive liquid from the liquid storage tank **200** into the clamping component **100**, so as to perform erosion-corrosion on the GDL **800**.

(12) In an embodiment, a water bath **230** is provided outside the liquid storage tank **200**.

(13) Referring to FIG. 2, the clamping component **100** includes a plurality of clamping plates **130** arranged in parallel. A middle of the clamping component **100** is provided with a flow channel. Both ends of the clamping component **100** are channel-free. The GDL **800** is provided between two adjacent clamping plates among the plurality of clamping plates **130**. Each of the plurality of clamping plates **130** is provided with a first water inlet **110** and a first water outlet **120**. The first water inlet **110** is connected to the first pipeline, and the first water outlet **120** is connected to a third pipeline connected to the liquid storage tank. The first water inlet **110** and the first water outlet **120** are configured such that the corrosive liquid enters the flow channel through the first water inlet **110** to perform the erosion-corrosion on the GDL **800**, and flows back to the liquid storage tank **200** through the first water outlet **120**.

(14) Specifically, a filtering tank **700** is provided between a second water inlet **210** of the liquid storage tank **200** and the first water outlet **120**. The corrosive liquid from the clamping component **100** passes through the filtering tank **700** for removing impurity and then flows back to the liquid storage tank **200**.

(15) A front end of the first water inlet **110** is connected to a float-type flowmeter **400** and a pulse valve **500** in sequence. The pulse valve **500** is connected to the peristaltic pump **300**.

(16) Specifically, the float-type flowmeter **400** is configured to accurately measure a flow rate of a water inflow.

(17) The pulse valve **500** (optionally opened) is configured to simulate a non-uniform flow under an operating condition of a fuel cell.

(18) The plurality of clamping plates **130** have the same size, and are each provided with a positioning hole **140** at the same position. The plurality of clamping plates **130** are configured to be fixed via a screw **150** passing through the positioning hole **140**. A heating component **600** is provided on an outer side of the clamping component **100**. In an embodiment, the heating component **600** is arranged on an outer side of an outermost clamping plate among the plurality of clamping plates **130**. In an embodiment, the heating component **600** is a heating sheet.

(19) Referring to FIG. 1, a thermocouple **160** is provided inside the plurality of clamping plates **130**, and is configured to measure an internal temperature of the clamping component **100** (0-100° C.).

(20) In an embodiment, the plurality of clamping plates **130** includes a combination of at least two of a blind plate without flow channels on both side surfaces, an end plate with a flow channel on

one side surface, and a middle plate with flow channels on both side surfaces. The flow channels are configured such that the liquid in each flow channel flows independently.

(21) In an embodiment, the flow channel is a serpentine groove, a parallel in-line groove, a wave-shaped groove or a dot-matrix groove. When flushing the GDL, the GDL is in fit with the groove, and the corrosive liquid flows through the groove to flush the GDL.

(22) In an embodiment, the plurality of clamping plates **130** are configured as a combination of an end plate and a blind plate. A flow-channel side surface of the end plate is opposite to the blind plate, with a GDL **800** sandwiched therebetween. A microporous-layer side surface of the GDL **800** is opposite to the flow-channel side surface of the end plate, such that the microporous layer of the GDL **800** can be independently flushed.

(23) In an embodiment, the plurality of clamping plates **130** are configured as a combination of two end plates and at least one middle plate. The at least one middle plate is located between the flow-channel side surfaces of the two end plates. The GDL **800** is sandwiched between the middle plate and the end plate, so as to simultaneously perform the erosion-corrosion on a microporous layer and a backing layer of the GDL **800**.

(24) In an embodiment, the GDL **800** is arranged at a middle of the flow-channel plate, and a gasket **170** is provided around the GDL **800** for sealing. A thickness of the gasket **170** is adjustable, such that the compression ratio of the GDL **800** can be adjusted by changing the thickness of the gasket **170**, thereby simulating the pressure environment of the GDL **800** when assembled into a fuel cell. In some embodiments, the thickness of the gasket **170** is 100-240 μm , and the gasket **170** is made of polytetrafluoroethylene (PTFE).

(25) In an embodiment, the corrosive liquid is selected from the group consisting of water, Fenton's reagent, ethanol, acetone, hydrogen peroxide, sulfuric acid, hydrochloric acid, nitric acid, sodium hydroxide and a combination thereof.

(26) In an embodiment, the liquid storage tank **200** includes a blue-capped mobile phase bottle with a height of 100-300 mm.

(27) A method for testing GDL durability using the above system is also provided, which includes the following steps.

(28) Step (1) A plurality of GDLs **800** are provided. The plurality of GDLs are identical to each other.

(29) Step (2) The plurality of GDLs **800** are respectively clamped between the plurality of clamping plates **130**.

(30) Step (3) The peristaltic pump **300** is turned on to perform erosion-corrosion on the plurality of GDLs **800**. N GDLs among the plurality of GDLs are taken out every M days.

(31) Step (4) The physical and chemical parameters of the plurality of GDLs and an uncorroded GDL are determined.

(32) The physical and chemical parameters include planar conductivity, machine-direction (MD) stiffness and weight loss rate.

Embodiment 1

(33) Referring to FIG. 3, a clamping component **100** is provided. Clamping plates **130** of the clamping component **100** are respectively a front end plate **141**, a first middle plate **142**, a second middle plate **143**, a third middle plate **144**, a fourth middle plate **145**, a fifth middle plate **146**, a sixth middle plate **147**, a seventh middle plate **148**, an eighth middle plate **149**, a ninth middle plate **1410** and a rear end plate **1411**.

(34) The front end plate **141** is provided with a water inlet a1 and a water outlet b1.

(35) The first middle plate **142** is provided with a water inlet a2 and a water outlet b2.

(36) The second middle plate **143** is provided with a water inlet a3 and a water outlet b3.

(37) The third middle plate **144** is provided with a water inlet a4 and a water outlet b4.

(38) The fourth middle plate **145** is provided with a water inlet a5 and a water outlet b5.

(39) The fifth middle plate **146** is provided with a water inlet a6 and a water outlet b6.

- (40) The sixth middle plate **147** is provided with a water inlet **a7** and a water outlet **b7**.
- (41) The seventh middle plate **148** is provided with a water inlet **a8** and a water outlet **b8**.
- (42) The eighth middle plate **149** is provided with a water inlet **a9** and a water outlet **b9**.
- (43) The ninth middle plate **1410** is provided with a water inlet **a10** and a water outlet **b10**.
- (44) The rear end plate **1411** is provided with a water inlet **a11** and a water outlet **b11**.
- (45) The water inlets and the water outlets are all $\frac{1}{8}$ thread to $\frac{1}{4}$ quick plugs.
- (46) The flow channels of the plates are all arranged in parallel.
- (47) All GDLs **800** are the same in model, and have a thickness of 200 μm .
- (48) All screws **150** are M5 screws with a torque of 10 N.Math.m.
- (49) A thickness of a PTFE gasket **170** is 140 μm (the thickness of the gasket should be less than that of the GDLs **800**).
- (50) The liquid pipelines are $\frac{1}{4}$ -PTFE hard pipes. The pipe at the peristaltic pump **300** is a silicone hose (hose interface 1:1, inner diameter 4.8 mm and wall thickness 1.6 mm), and a hard pipe-hose joint is a $\frac{1}{4}$ quick-plug to pagoda joint.
- (51) The peristaltic pump **300** is a Baoding Qili® BT100-02 driver with a YZ1515 pump head, and has a rotation speed of 100 rpm. A main path of the peristaltic pump **300** adopts a first 1:11 liquid distribution row with 11 branches for supplying the 11 water inlets.
- (52) The return water from the 11 water outlets is converged into a single path through a second 1:11 liquid distribution row and returned to the liquid storage tank **200**.
- (53) The liquid storage tank **200** is a blue-capped mobile phase bottle. The blue-capped mobile phase bottle has a height of 100-300 mm, such as 250 mm. The blue-capped mobile phase bottle has a diameter of 50-150 mm, such as 100 mm. A cap of the blue-capped mobile phase bottle has a first water inlet **110** and a first water outlet **120**. The first water outlet **120** extends inwardly to a bottom of the blue-capped mobile phase bottle by means of a hose, and the first water inlet **110** extends inwardly above a liquid level by means of a hose.
- (54) **S100**. 10 GDL samples are cut using a GDL special cutting machine.
- (55) **S200**. 500 mL of a Fenton's reagent is prepared as a corrosive liquid. The corrosive liquid is loaded into the liquid storage tank **200**. The 10 GDL samples are respectively clamped between the clamping component **100**, where one GDL sample is clamped between every two adjacent clamping plates **130**. The specific clamping conditions are described as follows.
- (56) Sample No. 1 is clamped between the front end plate **141** and the first middle plate **142**.
- (57) Sample No. 2 is clamped between the first middle plate **142** and the second middle plate **143**.
- (58) Sample No. 3 is clamped between the second middle plate **143** and the third middle plate **144**.
- (59) Sample No. 4 is clamped between the third middle plate **144** and the fourth middle plate **145**.
- (60) Sample No. 5 is clamped between the fourth middle plate **145** and the fifth middle plate **146**.
- (61) Sample No. 6 is clamped between the fifth middle plate **146** and the sixth middle plate **147**.
- (62) Sample No. 7 is clamped between the sixth middle plate **147** and the seventh middle plate **148**.
- (63) Sample No. 8 is clamped between the seventh middle plate **148** and the eighth middle plate **149**.
- (64) Sample No. 9 is clamped between the eighth middle plate **149** and the ninth middle plate **1410**.
- (65) Sample No. 10 is clamped between the ninth middle plate **1410** and the rear end plate **1411**.
- (66) The 10 GDL samples are in one-to-one correspondence with the gaskets, and a ratio of the gasket thickness to the GDL thickness is the assembly compression ratio of the GDL.
- (67) **S300**. The liquid storage tank **200** is placed in the water bath **230** at 80° C. The water inlet and the water outlet of each of the clamping plates **130** are respectively connected by means of hoses. The peristaltic pump **300** is connected to the liquid storage tank **200**. The peristaltic pump **300** is turned on to pump the corrosive liquid, so that the entire corrosive liquid forms a circulating loop.
- (68) **S400**. One sample was taken out every other day, rinsed with deionized water and dried at 100° C. After 10 days, all of the 10 GDL samples are taken out and determined for physical and chemical parameters, including weight loss rate (with the results shown in FIG. 4), surface

resistivity (with the results shown in FIG. 5), and MD direction stiffness (with the results shown in FIG. 6).

(69) As shown in FIG. 4, both side surfaces of the GDL are subjected to erosion-corrosion for 10 days using the Fenton's reagent. As the corrosion time increases, the material weight loss continuously increases. This is because the graphite filled on the backing layer and the carbon black on the microporous layer fall off under chemical corrosion and are lost under mechanical erosion of the water flow.

(70) As shown in FIG. 5, as the corrosion time increases, the surface resistivity of the GDL continuously increases. This may be because an important part of the GDL for conductivity, i.e., the graphite filled on the backing layer, falls off under chemical corrosion and is lost under the mechanical erosion of the water flow.

(71) As shown in FIG. 6, with the increase of corrosion time, the bending stiffness of the GDL continuously decreases until it almost loses stiffness on the tenth day, which is due to the loss of bonded carbon among the carbon fibers under chemical corrosion and mechanical stress, and insufficient bonding force among the carbon fibers, resulting in a significant decrease in the bending stiffness of the material.

(72) It should be noted that the specific parameters or some reagents in the embodiments are merely illustrative, and are not intended to limit the present disclosure. It should be understood that various modifications or replacements made by those skilled in the art without departing from the spirit of the disclosure shall fall within the scope of the present disclosure defined by the appended claims.

Claims

1. A system for testing durability of a gas diffusion layer (GDL), comprising: a clamping component; a liquid storage tank; and a peristaltic pump; wherein the clamping component is configured to fix the GDL, a middle of the clamping component is provided with a flow channel, and both ends of the clamping component are channel-free; the liquid storage tank is configured to store a corrosive liquid; the peristaltic pump is arranged between the clamping component and the liquid storage tank; and the peristaltic pump is connected to the clamping component via a first pipeline, and is connected to the liquid storage tank via a second pipeline; the peristaltic pump is configured to be turned on to pump the corrosive liquid from the liquid storage tank into the clamping component during testing, so as to perform erosion-corrosion on the GDL; the clamping component comprises a plurality of clamping plates arranged in parallel; the GDL is located between two adjacent clamping plates among the plurality of clamping plates; each of the plurality of clamping plates is provided with a water inlet and a water outlet; the water inlet is connected to the first pipeline, and the water outlet is connected to a third pipeline connected to the liquid storage tank; and the water inlet and the water outlet are configured such that the corrosive liquid enters the flow channel through the water inlet to perform the erosion-corrosion on the GDL, and flows back to the liquid storage tank through the water outlet; the plurality of clamping plates comprise two end plates with one flow-channel side surface and at least one middle plate with two flow-channel side surfaces; the at least one middle plate is located between the flow-channel side surfaces of the two end plates; the GDL is configured to be tested for physical and chemical parameters, wherein the physical and chemical parameters comprise planar conductivity, machine-direction (MD) stiffness and weight loss rate; and an end of the water inlet is connected to a float-type flowmeter and a pulse valve in sequence, and the pulse valve is connected to the peristaltic pump.

2. The system of claim 1, wherein the plurality of clamping plates have the same size, and are each provided with a positioning hole at the same position; and the plurality of clamping plates are configured to be fixed via a screw passing through the positioning hole.

3. The system of claim 1, wherein a gasket is provided around the GDL for sealing.
4. The system of claim 1, wherein the corrosive liquid is selected from the group consisting of water, Fenton's reagent, ethanol, acetone, hydrogen peroxide, sulfuric acid, hydrochloric acid, nitric acid, sodium hydroxide and a combination thereof.
5. The system of claim 1, wherein the liquid storage tank comprises a blue-capped mobile phase bottle with a height of 100-300 mm.
6. A system for testing durability of a GDL, comprising: a clamping component; a liquid storage tank; and a peristaltic pump; wherein the clamping component is configured to fix the GDL, a middle of the clamping component is provided with a flow channel, and both ends of the clamping component are channel-free; the liquid storage tank is configured to store a corrosive liquid; the peristaltic pump is arranged between the clamping component and the liquid storage tank; and the peristaltic pump is connected to the clamping component via a first pipeline, and is connected to the liquid storage tank via a second pipeline; the peristaltic pump is configured to be turned on to pump the corrosive liquid from the liquid storage tank into the clamping component during testing, so as to perform erosion-corrosion on the GDL; the clamping component comprises a plurality of clamping plates arranged in parallel; the GDL is located between two adjacent clamping plates among the plurality of clamping plates; each of the plurality of clamping plates is provided with a water inlet and a water outlet; the water inlet is connected to the first pipeline, and the water outlet is connected to a third pipeline connected to the liquid storage tank; and the water inlet and the water outlet are configured such that the corrosive liquid enters the flow channel through the water inlet to perform the erosion-corrosion on the GDL, and flows back to the liquid storage tank through the water outlet; the plurality of clamping plates comprise two end plates with one flow-channel side surface and at least one middle plate with two flow-channel side surfaces; the at least one middle plate is located between the flow-channel side surfaces of the two end plates; the GDL is configured to be tested for physical and chemical parameters, wherein the physical and chemical parameters comprise planar conductivity, machine-direction (MD) stiffness and weight loss rate; and a heating component is provided on an outer side of the clamping component.
7. A method for testing GDL durability using a system for testing durability of a GDL, the system comprising: a clamping component; a liquid storage tank; and a peristaltic pump; wherein the clamping component is configured to fix the GDL, a middle of the clamping component is provided with a flow channel, and both ends of the clamping component are channel-free; the liquid storage tank is configured to store a corrosive liquid; the peristaltic pump is arranged between the clamping component and the liquid storage tank; and the peristaltic pump is connected to the clamping component via a first pipeline, and is connected to the liquid storage tank via a second pipeline; the peristaltic pump is configured to be turned on to pump the corrosive liquid from the liquid storage tank into the clamping component during testing, so as to perform erosion-corrosion on the GDL; the clamping component comprises a plurality of clamping plates arranged in parallel; the GDL is located between two adjacent clamping plates among the plurality of clamping plates; each of the plurality of clamping plates is provided with a water inlet and a water outlet; the water inlet is connected to the first pipeline, and the water outlet is connected to a third pipeline connected to the liquid storage tank; and the water inlet and the water outlet are configured such that the corrosive liquid enters the flow channel through the water inlet to perform the erosion-corrosion on the GDL, and flows back to the liquid storage tank through the water outlet; the plurality of clamping plates comprise two end plates with one flow-channel side surface and at least one middle plate with two flow-channel side surfaces; the at least one middle plate is located between the flow-channel side surfaces of the two end plates; and the GDL is configured to be tested for physical and chemical parameters, wherein the physical and chemical parameters comprise planar conductivity, machine-direction (MD) stiffness and weight loss rate; the method comprising: (1) providing a plurality of GDLs, wherein the plurality of GDLs are identical to each other; (2) respectively clamping the plurality of GDLs between the plurality of clamping plates; (3)

turning on the peristaltic pump to perform erosion-corrosion on the plurality of GDLs, and taking out N GDLs among the plurality of GDLs every M days; and (4) determining the physical and chemical parameters of the plurality of GDLs and an uncorroded GDL.
